

The empirical recurrence for OEIS A255145: recovering a conjecture that is not written down

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Abstract

OEIS A255145 records “Empirical recurrence of order 76”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 80$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 76, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 76 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A255145 is “Number of $(n+2) \times (4+2)$ 0..1 arrays with no 3×3 subblock diagonal sum 0 and no antidiagonal sum 0 and no row sum 1 and no column sum 1.” It has offset 1 and begins

784, 4576, 29731, 190731, 1241971, 8173810, 53762469, 353508918, ...

The entry states:

Empirical recurrence of order 76 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 11:38:55, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1914$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 80$ of the 1914, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 80$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 160$ exact terms of the model returns order 76 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{76} c_i a(n-i),$$

c_1	5	c_{20}	162600038	c_{39}	−27734688425	c_{58}	291850215
c_2	5	c_{21}	398605209	c_{40}	−7878847991	c_{59}	−105076206
c_3	86	c_{22}	299728823	c_{41}	18992372243	c_{60}	−180988013
c_4	−146	c_{23}	−433643217	c_{42}	32202769038	c_{61}	−97591309
c_5	−979	c_{24}	−1415720346	c_{43}	23223954918	c_{62}	−15938954
c_6	−2643	c_{25}	−1472275291	c_{44}	−441443568	c_{63}	16717834
c_7	830	c_{26}	370938986	c_{45}	−20418308559	c_{64}	15123914
c_8	31120	c_{27}	3439905869	c_{46}	−23418175725	c_{65}	5796118
c_9	72561	c_{28}	4905066585	c_{47}	−11161229102	c_{66}	−56492
c_{10}	−11229	c_{29}	2002634231	c_{48}	4949160558	c_{67}	−1224058
c_{11}	−491986	c_{30}	−4973105229	c_{49}	12728907076	c_{68}	−612268
c_{12}	−1172995	c_{31}	−10928113888	c_{50}	10228175512	c_{69}	−143528
c_{13}	−101007	c_{32}	−9671183640	c_{51}	2350485220	c_{70}	62664
c_{14}	5414867	c_{33}	1265555282	c_{52}	−3476112990	c_{71}	27088
c_{15}	11540991	c_{34}	15219111933	c_{53}	−4641102276	c_{72}	6696
c_{16}	3052797	c_{35}	21532298419	c_{54}	−2420293052	c_{73}	−200
c_{17}	−36517085	c_{36}	11692078687	c_{55}	40901875	c_{74}	−1456
c_{18}	−80711624	c_{37}	−9418854814	c_{56}	1132983687	c_{75}	48
c_{19}	−39106828	c_{38}	−27915138627	c_{57}	915679287	c_{76}	96

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 76, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $76 + 4$ terms removed returns order 76 as well, so $\deg q_{\min} = 76 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $76 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 76 that this sequence satisfies — and that family turns out to have exactly one member.

References

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